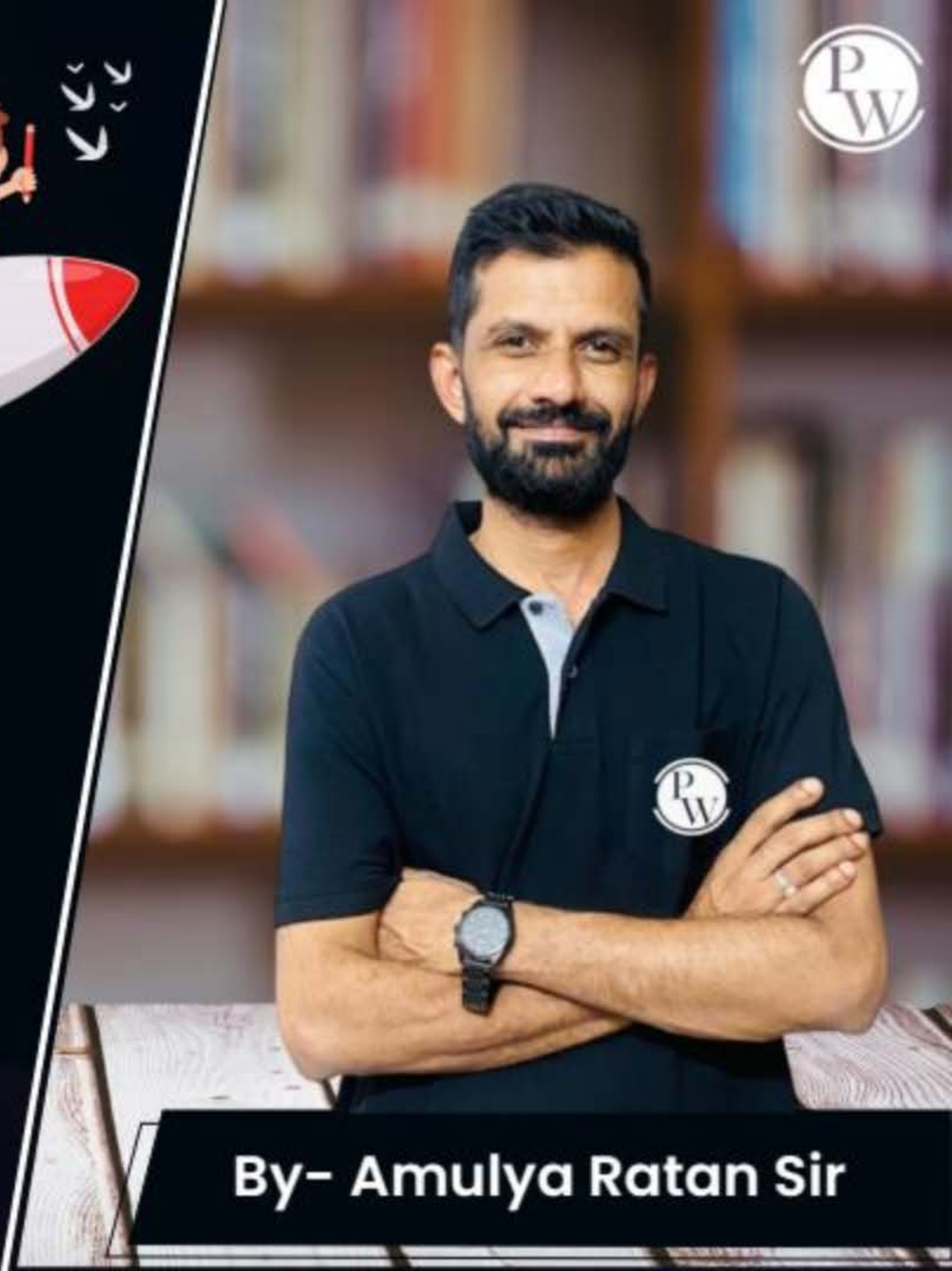
GATE ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 07



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Interest

Topics to be Covered

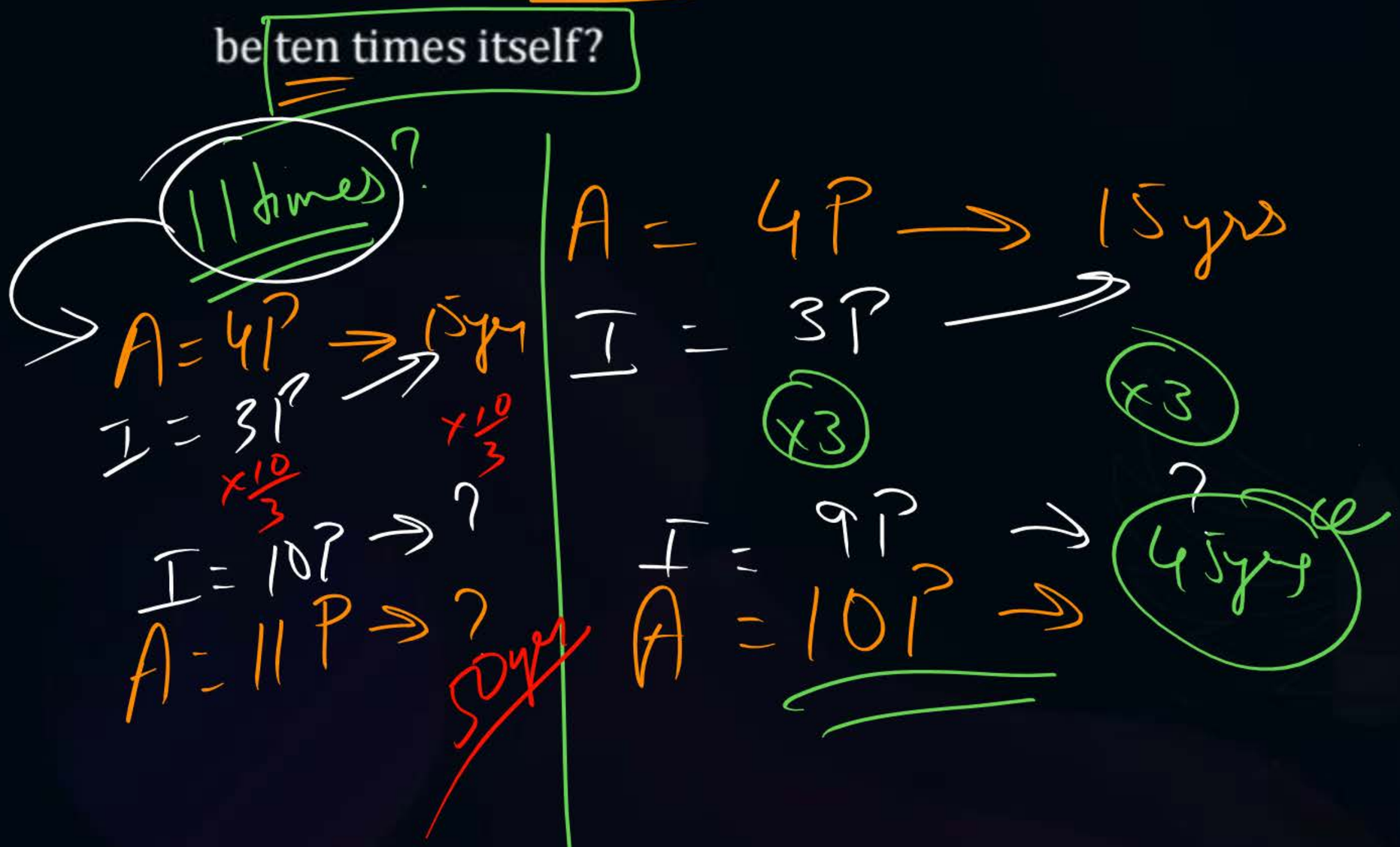


Topic-1 Sequence & Data Sufficiency

[MCQ]

$15 \times \frac{10}{3} = 50$ yrs

#Q. An investment quadruple itself in 15 years. Then in how many years it would be ten times itself?



[MCQ]

16P?

#Q. An investment doubles itself in 6years on compound interest. Then in how many years it would be sixteen times itself?

$A = P\left(1+\frac{R}{100}\right)^n$

$2P = P\left(1+\frac{R}{100}\right)^6$

$(2P)^4 = P\left(1+\frac{R}{100}\right)^{6\times 4}$

2^1 2P ⇒ 6yrs ×1 ✓

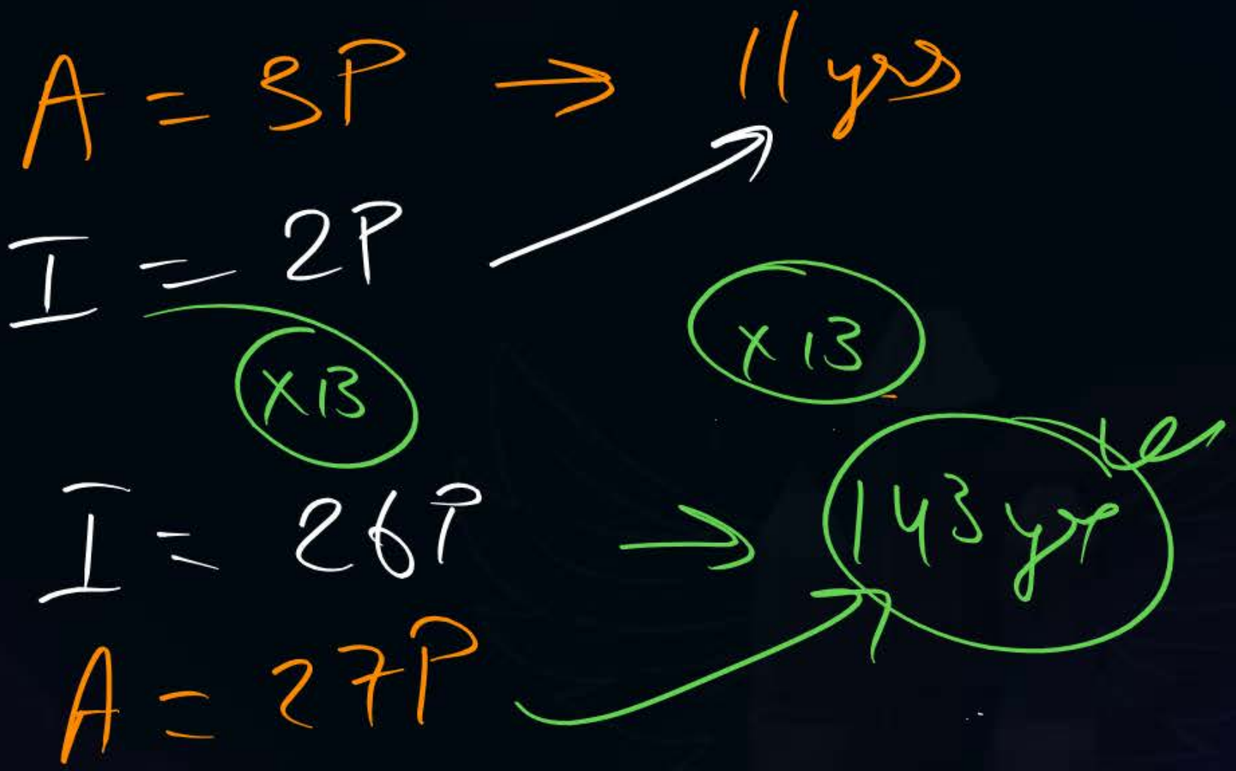
2^2 4P ⇒ 12yrs ×2 ✓

2^3 8P ⇒ 18yrs ×3 ✓

2^4 16P ⇒ 24yrs ×4 ✓

[MCQ]

#Q. An investment triples itself in 11years. Then in how many years it would be twenty seven times itself?



[MCQ]

#Q. A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

16×12000

$2P = 5 \text{ yrs}$

$= 1,92,000$

$(\quad)^4$ — $\times 4$

$16P = 20 \text{ yrs}$

- A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

- Sol: The rate of interest at which ₹12000 doubles after 5 years is given by
- $R = 100(x^{\frac{1}{5}} - 1)$
- $= 100(2^{\frac{1}{5}} - 1)$
- $=100 \times (1.1486 - 1)$
- $= 100 \times 0.1486 = 14.86\%$

Therefore, after 20 years it becomes,

$A = ₹12000(1 + 14.86/100)^{20} = ₹12000\ (1.1486)^{20}$

$= ₹12000 \times 15.97 = ₹\ 191671.474$ (approx.)

[MCQ]

$104\% = \frac{104}{100} = 1.04$

#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

Assignment

A ₹ 180

B ₹ 210

C ₹ 204

D ₹ 220

$2500 \times 1.04 \times 1.04$

$= 2704$

$2704 - 2500$

$= 204$

$CI = P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$

$= 2500\left[\left(1+\frac{4}{100}\right)^2 - 1\right]$

$= 204$

[MCQ]

$\frac{4000}{12} = P$

#Q. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched ₹480 more. Find the principal.

$SI_1 - SI_2 = 480$

Assignment

A ₹3000

B ₹4000

C ₹5000

D ₹6000

$$480 = \frac{P \times (R+4) \times 3}{100} - \frac{P \times R \times 3}{100}$$

$$480 = \frac{3P(R+4) - 3P(R)}{100} = \frac{3P(R+4-R)}{100} = \frac{12P}{100}$$

[MCQ]

#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

$$A = P\left(1 + \frac{TR}{100}\right)$$

$$2P = P\left(1 + \frac{T \times 6}{100}\right)$$

$$1 = \frac{6T}{100}$$

$$T = \frac{100}{6} = \frac{50}{3} = 16\frac{2}{3}\text{ yr}$$

A. 6 ⅓ years

B. 16 ⅔ years

C. 33 ⅓ years

D. 21 years

[MCQ]

#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?

Assignment

A. 1 ⅓ years

B. 1 ½ years

C. 2 ⅓ years

D. 2 ½ years

$A = P\left(1+\frac{r}{100}\right)^n$

$926.10 = 800\left(1+\frac{5}{100}\right)^{2n}$

$\Rightarrow \frac{92610}{80000} = \left(\frac{21}{20}\right)^{2n}$

$\Rightarrow \frac{9261}{8000} = \left(\frac{21}{20}\right)^{2n}$

$\left(\frac{21}{20}\right)^3 = \left(\frac{21}{20}\right)^{2n}$

$3 = 2n$

$n = \frac{3}{2} = 1\frac{1}{2}$ years

[MCQ]

$n = 8$ (Annual) $T = 2$ yrs (Annual)
$n = 4$ (Half) ₹ 24.64

#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

A ₹ 21.25

B ₹ 24.64

C ₹ 27.85

D ₹ 29.94

41 ⇒ 2500

$0.04 \times 2500 = 100$

400

$Diff = P\left(\frac{r}{100}\right)^2$

$16 = P\left(\frac{8}{100}\right)^2$

$16 = P \times \frac{64}{10000}$

$P = \frac{10000}{4} = 2500$

Assignment

$A = 2500 \times 1.04 \times 1.04 \times 1.04 \times 1.04$

$A = 2924.\overline{64}$

$I = 2924.64 - 2500$

[MCQ]

#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

P T C A

= ?

A. $7\frac{1}{7}\%$

B. $9\frac{2}{3}\%$

C. $3\frac{1}{3}\%$

D. $6\frac{1}{3}\%$

$$A = P\left(1+\frac{r}{100}\right)^n$$

$$\frac{6750}{5488} = \left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \frac{3375}{2744} = \left(1+\frac{r}{100}\right)^3$$

Assignment

$$\left(\frac{15}{14}\right)^3 = \left(1+\frac{r}{100}\right)^3$$

$$\frac{15}{14} = 1+\frac{r}{100}$$

$$\frac{1}{14} = \frac{r}{100}$$

$$r = \frac{100}{14} = \frac{50}{7} = 7\frac{1}{7}\%$$

Series & Sequence

01, 04, 09, 16, 25, 36, 49, 64 — Number Series



10, 40, 90, 61, 52, 63, 94, ___

[MCQ]

#Q. Find the missing number in the series: 2, 3, 5, 7, 11, ____, 17

13

A 13

B 14

C 9

D 15

[MCQ]

#Q. Find the next number in the series: 10, 12, 16, 24, 40, 72, 136, ___



$2^7 = 128$

A 272

B 382

C 264

D 292

136
128
264

[MCQ]

#Q. Find the missing number in the series: 979, 330, ____, 40, 15, 6

A 113

B 89

C 96

D 114

339
-9
330

990
-11
979

6 × 3 (-3) = 15
15 × 3 (-5) = 40
40 × 3 (-7) = 113
113 × 3 (-9) = 330
330 × 3 (-11) = 979

[MCQ]

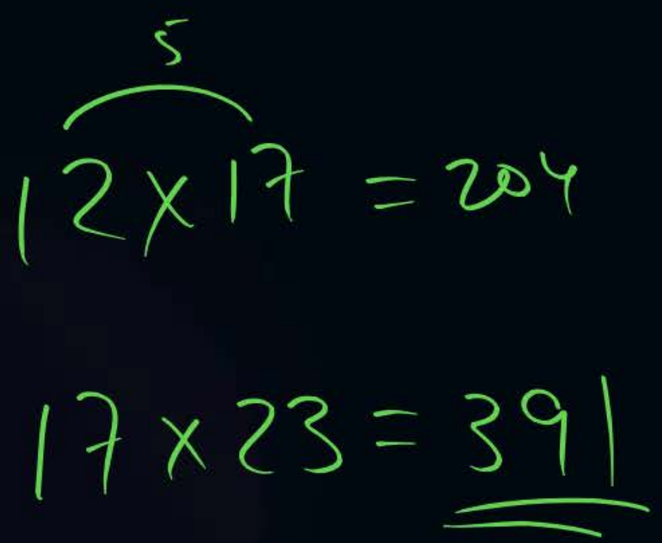
#Q. Find the missing number in the series: 391, 204, ____, 40, 15, 6

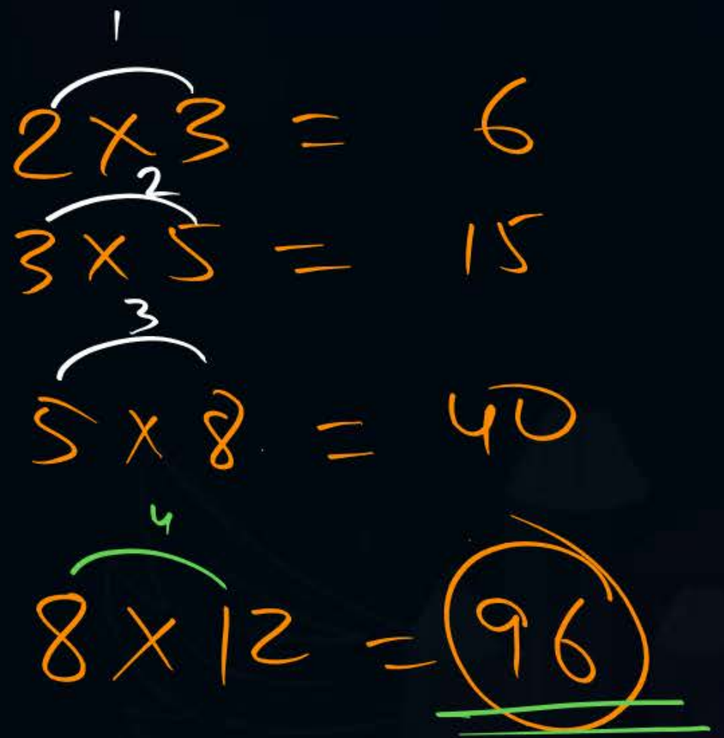
A 96

B 112

C 89

D 100





[MCQ]

#Q. Find the next number in the series: 1, 2, 10, 37, _____

Assignment

A 100

B 101

C 102

D 103

[MCQ]

#Q. Find the next number in the series: 2, 2, 4, 12, 48, _____

Assignment

A 96

B 250

C 240

D 215

[MCQ]

#Q. Find the next number in the series: 13, 17, 21, 25, _____

Assignment

A 27

B 28

C 29

D 31

[MCQ]

#Q. Find the next number in the series: 43, 44, 48, 57, 73, ____

Assignment

A 98

B 97

C 87

D 99

[MCQ]

#Q. Find the next number in the series: 1, 2, 5, 10, 17, 26, 37, _____

Assignment

A 51

B 50

C 49

D 65

[MCQ]

#Q. Find the missing number in the series: 21, 32, 45, 60, ____, 96, 117, 140

Assignment

A 75

B 74

C 73

D 77

DATA SUFFICIENCY

Please read the directions given carefully before answering the questions.

Directions



DATA SUFFICIENCY



Please read the directions given carefully before answering the questions.

Select option 'A', if statement (1) is sufficient to get the answer of given question.

Select option 'B', if statement (2) is sufficient to get the answer of given question.

Select option 'C', if statement (1) & (2) TOGETHER are sufficient to get the answer.

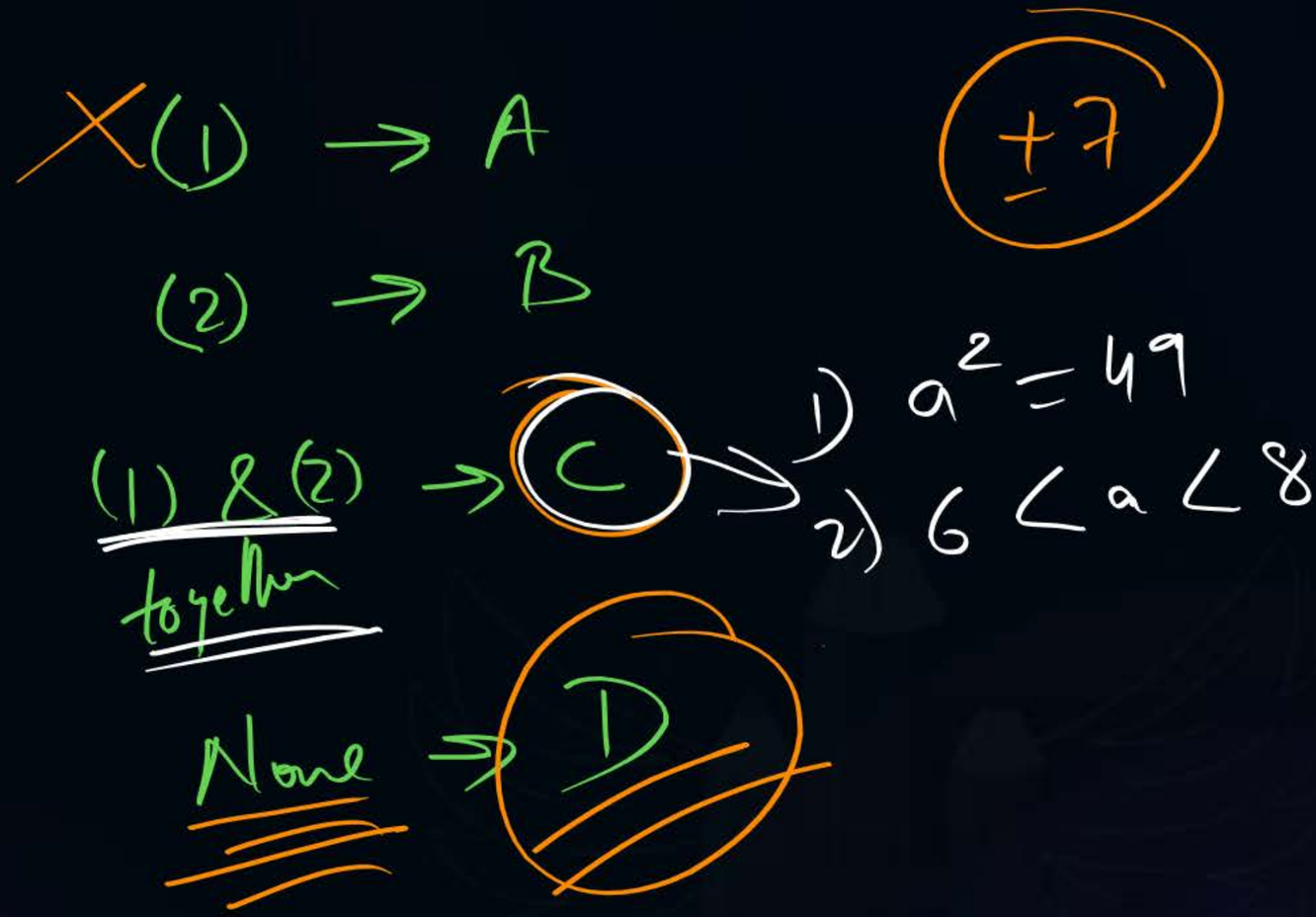
Select option 'D', if NONE of the statement is sufficient to get the answer.

[MCQ]

#Q. What is the value of 'a'?

1) $a^2 = 49$

2) $16 < a < 18$



[MCQ]

#Q. Is 'x' greater than 49?

1) $x < -8$

2) $x > 1000$

Directions

(1) → A

(2) → B

(1) & (2) → C ?

To Both

None → D

(1) OR (2)

NO

YES

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. How much is 20 percent of a certain number?

1) 10 percent of the number is 5.

2) 40 percent of twice the number is 40.

40% of 2n = 40

D

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

B.P : Flour

#Q. A thoroughly blended biscuit mix includes only flour and baking powder.

What is the ratio of the number of grams of baking powder to the number of grams of flour in the mix?

0.1 : 9.9

1 : 99

1) Exactly 9.9 grams of flour is contained in 10 grams of the mix.

2) Exactly 0.3 grams of baking powder is found in 30% of the mix.

30g 100g

1000g

300g

B.P : Flour

0.3 : 29.7

0.3 : 299.7

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. Is 'r' greater than 0.27?

1) r is greater than ¼.

2) r is equal to 3/10.

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. What is the value of the sum of a list of n odd integers?

Assignment

1) n = 8.

2) The square of the number of integers on the list is 64.

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

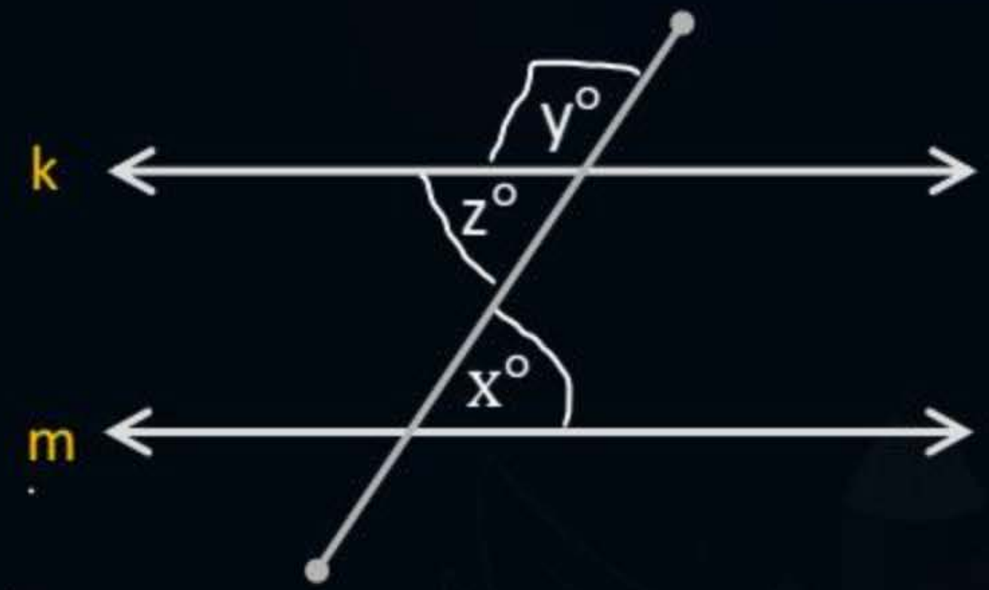
[MCQ]

#Q. In the given figure, if lines k and m are parallel, what is value of x?

Assignment

1) y = 120.

2) z = 60



Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. If r and s are positive integers, r is what percent of s?

1) r = 3/4 s

2) r ÷ s =75/100

Assignment

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. If $5x + 3y = 17$, what is the value of x?

1) x is a positive integer.

2) $y = 4x$

Assignment

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. Is the integer 'k' a prime number?

1) $2k = 16$.

2) $1 < k < 4$

Assignment

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. Water is pumped into a partially filled tank at a constant rate through an inlet pipe. At the same time, water is pumped out of the tank at a constant rate though an outlet pipe. At what rate, in gallons per minute, is the amount of water in the tank increasing?

Assignment

1) The amount of water initially in the tank is 200gallons.

2) Water is pumped into the tank at a rate of 10 gallons per minute and out of the tank at rate of 10 gallons every 2 ½ minutes.

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient

[MCQ]

#Q. What number is 15 percent of x?

1) 18 is 6 percent of x.

2) 2/3 of x is 200.

Assignment

Give your answers as per the following statements

A : Statement (1) alone is sufficient but statement (2) alone is not sufficient

B : Statement (2) alone is sufficient but statement (1) alone is not sufficient

C : Both statements together are sufficient but neither statement alone is sufficient

D : Each statement alone is sufficient

E : Both statement together are still not sufficient



2 mins Summary

Topic Sequence & Data Sufficiency

THANK - YOU